

Scenario audit of Cuba's population decline, 2021–2026

Primary scenario envelope: ~1.97–2.79 million of loss in 2021–2025. Illustrative central scenario: ≈8.58 M and **about 1 in 5** on the official 2021 base. Against the **official 2021** headcount, the gap toward 2026 approaches **1 in 4** — the same quantity, not a second definition. Emigration accounts for ≈92% of the gap under the central scenario.

1.97–2.79 M

across-scenario envelope
(primary uncertainty)

8.58 M

central scenario end-2025
(not an identified estimator)

2.0×

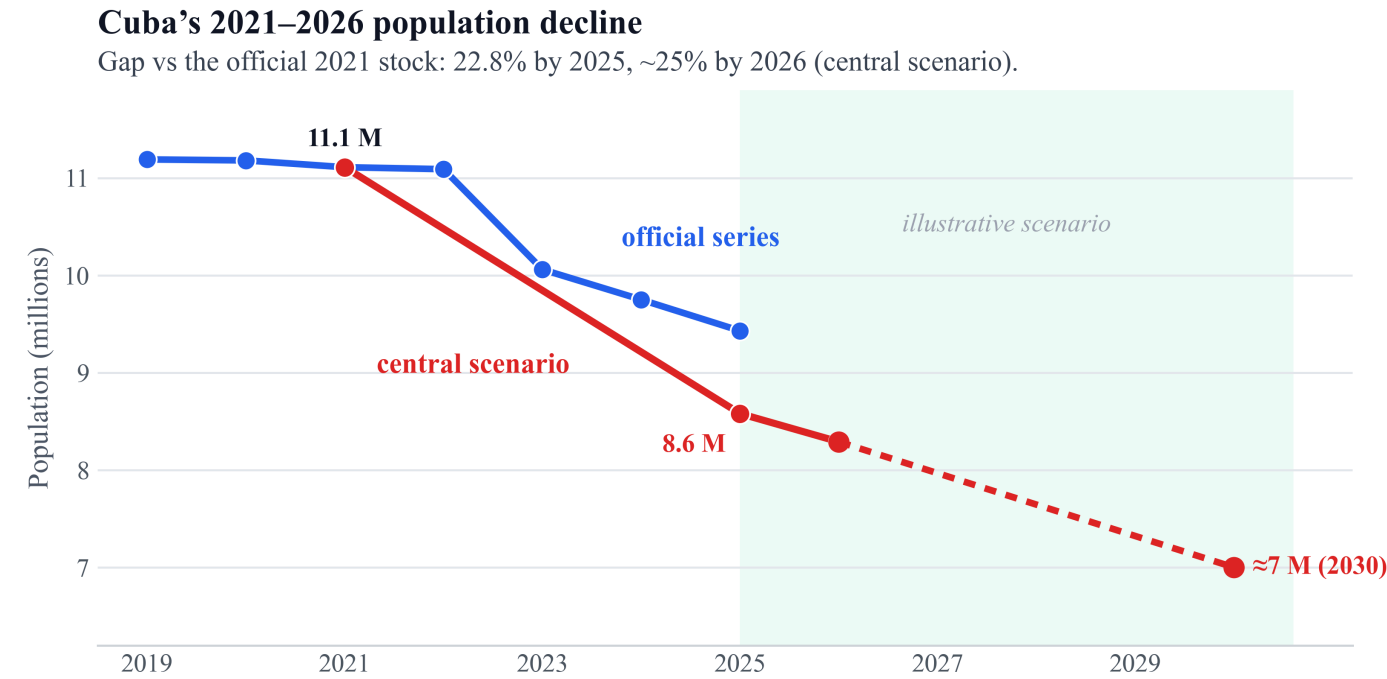
deaths roughly double births
(136k vs 68k in 2025)

1.05 M

settled-destination floor
(U.S. ~800k + non-U.S. ~250k)

Official headcount versus scenarios

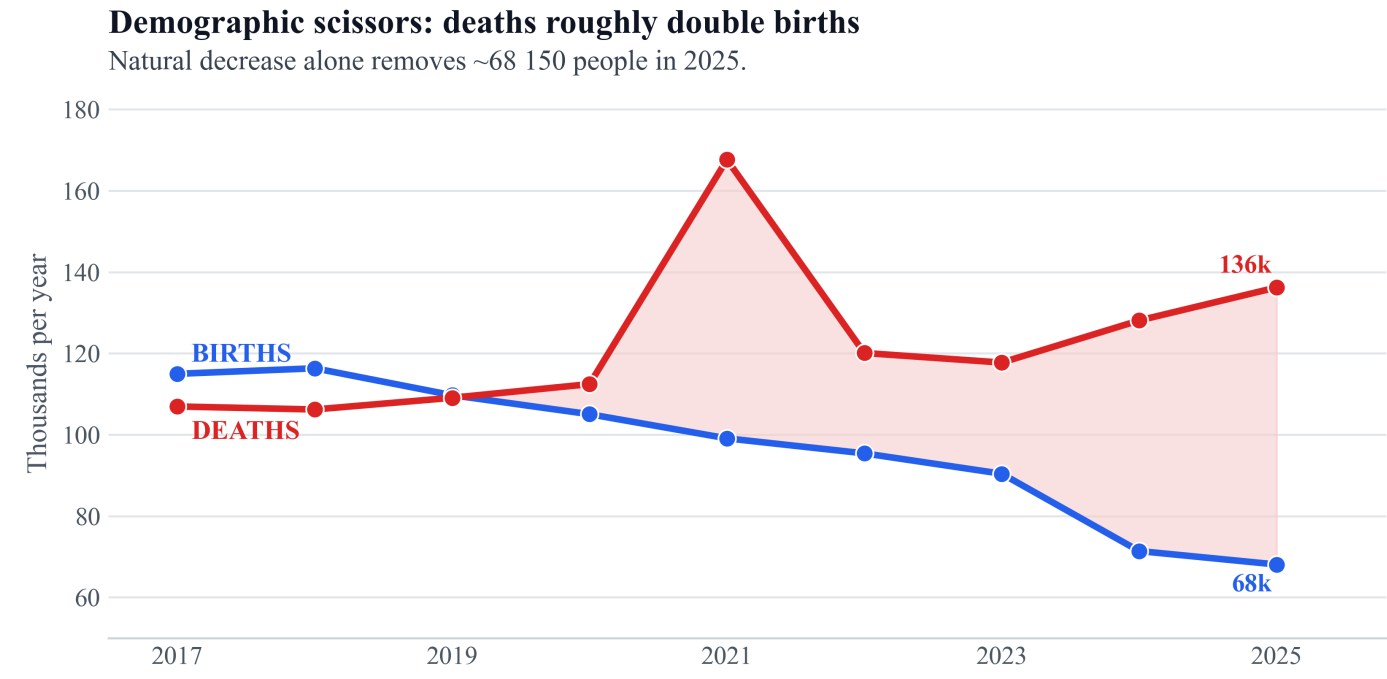
ONEI ≈9.43 M vs the central scenario ≈8.58 M (end-2025). Post-2026: illustrative trend, not a forecast.



The post-2026 segment is a non-probabilistic trend scenario, not a forecast.
Source: ONEI / Monte Carlo scenarios (central scenario = illustrative central path).

Demographic scissors: more deaths, fewer births

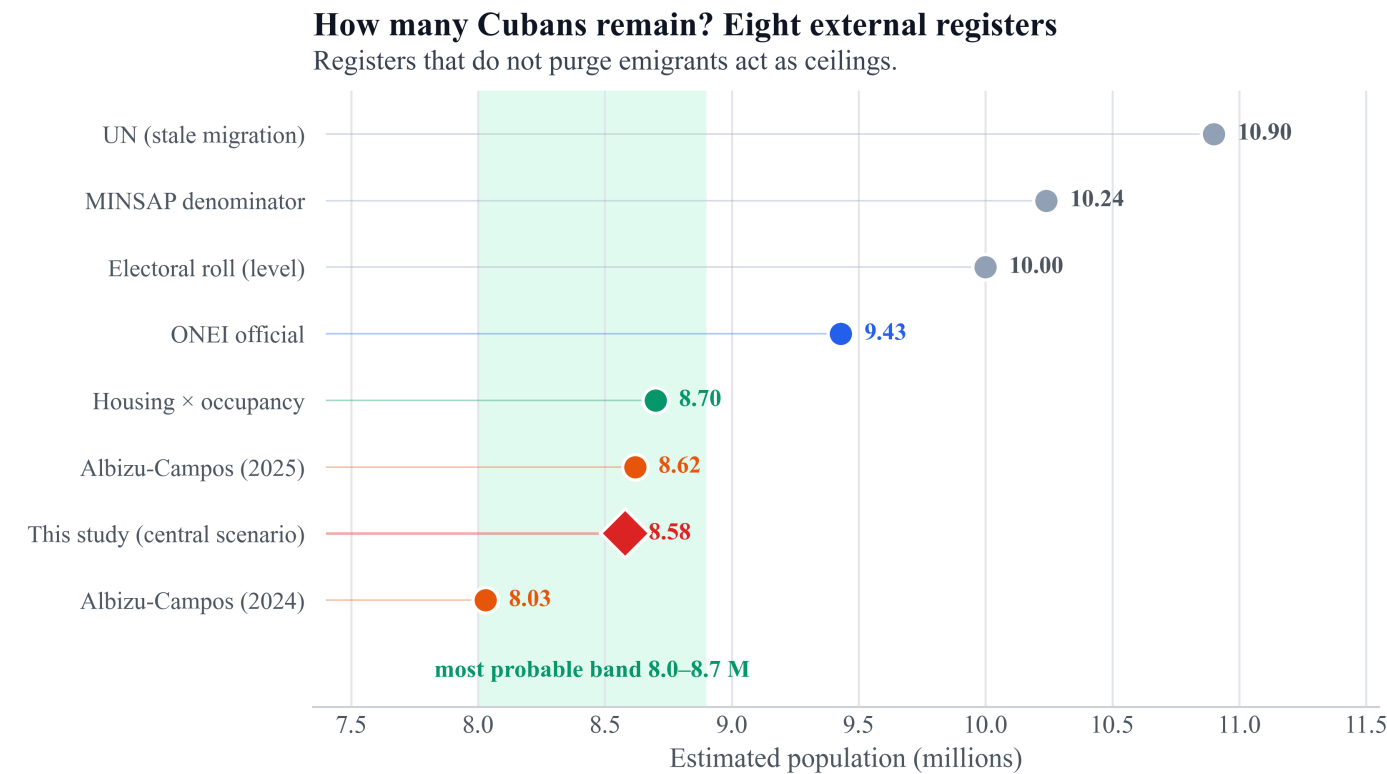
Since 2020 deaths exceed births by a wide margin. In 2025 two people die for every baby born.



Source: ONEI demographic yearbooks. The 2021 spike corresponds to the COVID wave.

Eight sources, soft band 8.0–8.7 M

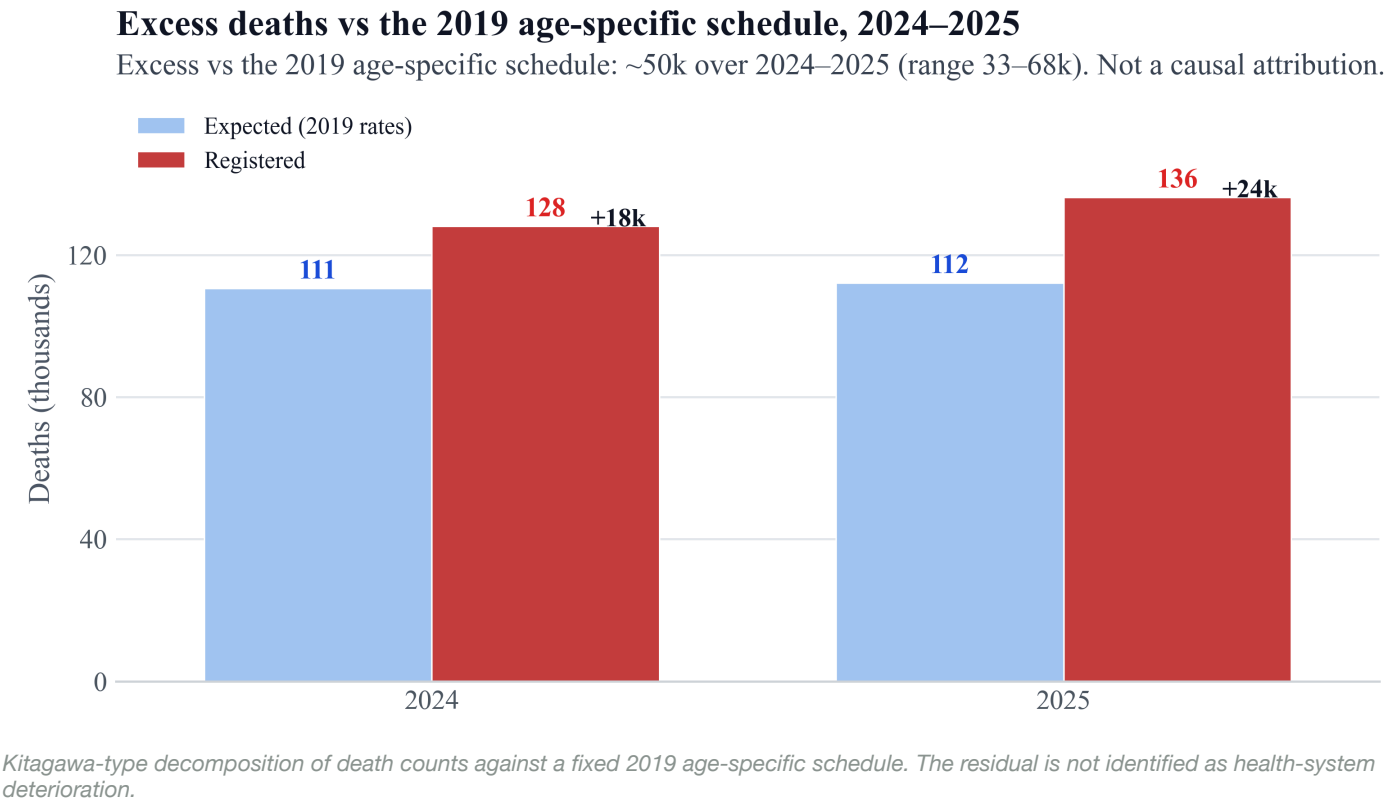
High ceilings (UN, electoral roll) vs exit-correcting anchors (Albizu, housing). The central scenario is the study estimand, not external validation.



Electoral roll, MINSAP, housing occupancy, UN, Albizu-Campos (2024, 2025), and the central scenario.

Excess deaths vs the 2019 schedule, 2024–2025

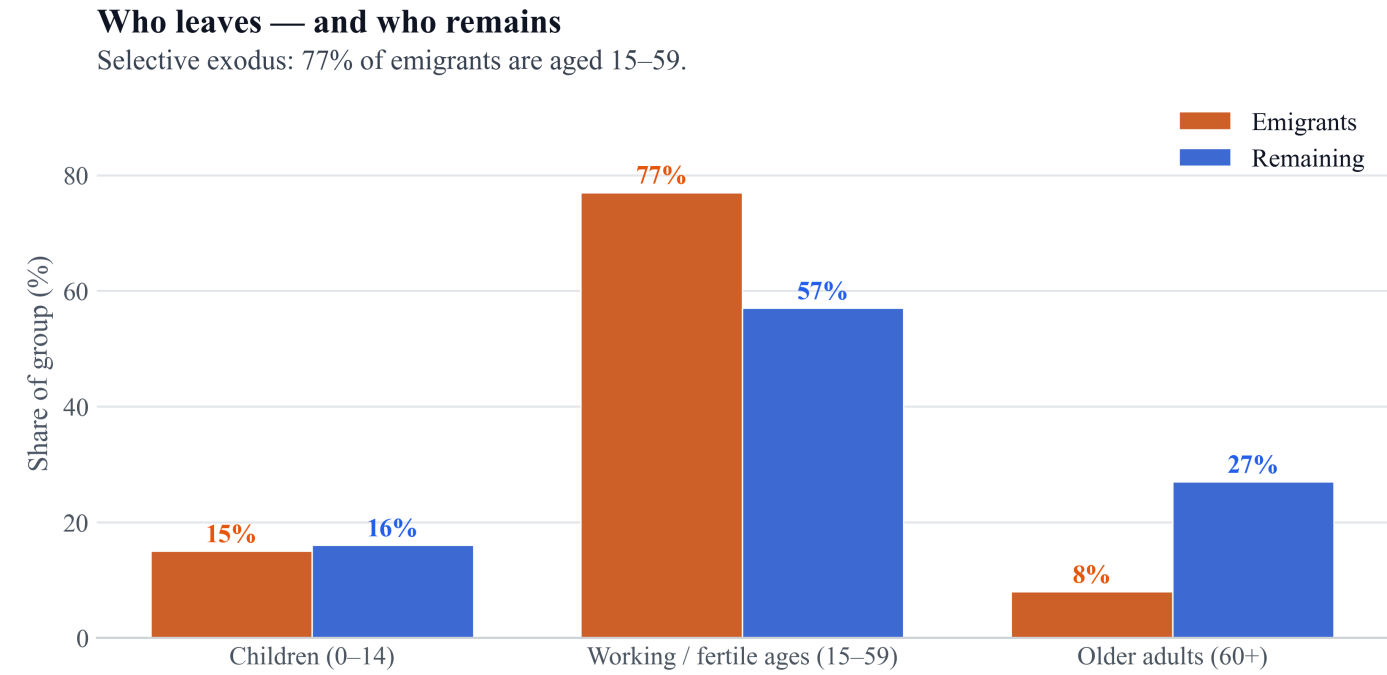
Illustrative ~28.5k/yr vs 2019 schedule (band 18.6–38.2k).



Kitagawa-type decomposition of death counts against a fixed 2019 age-specific schedule. The residual is not identified as health-system deterioration.

The young leave; the old remain

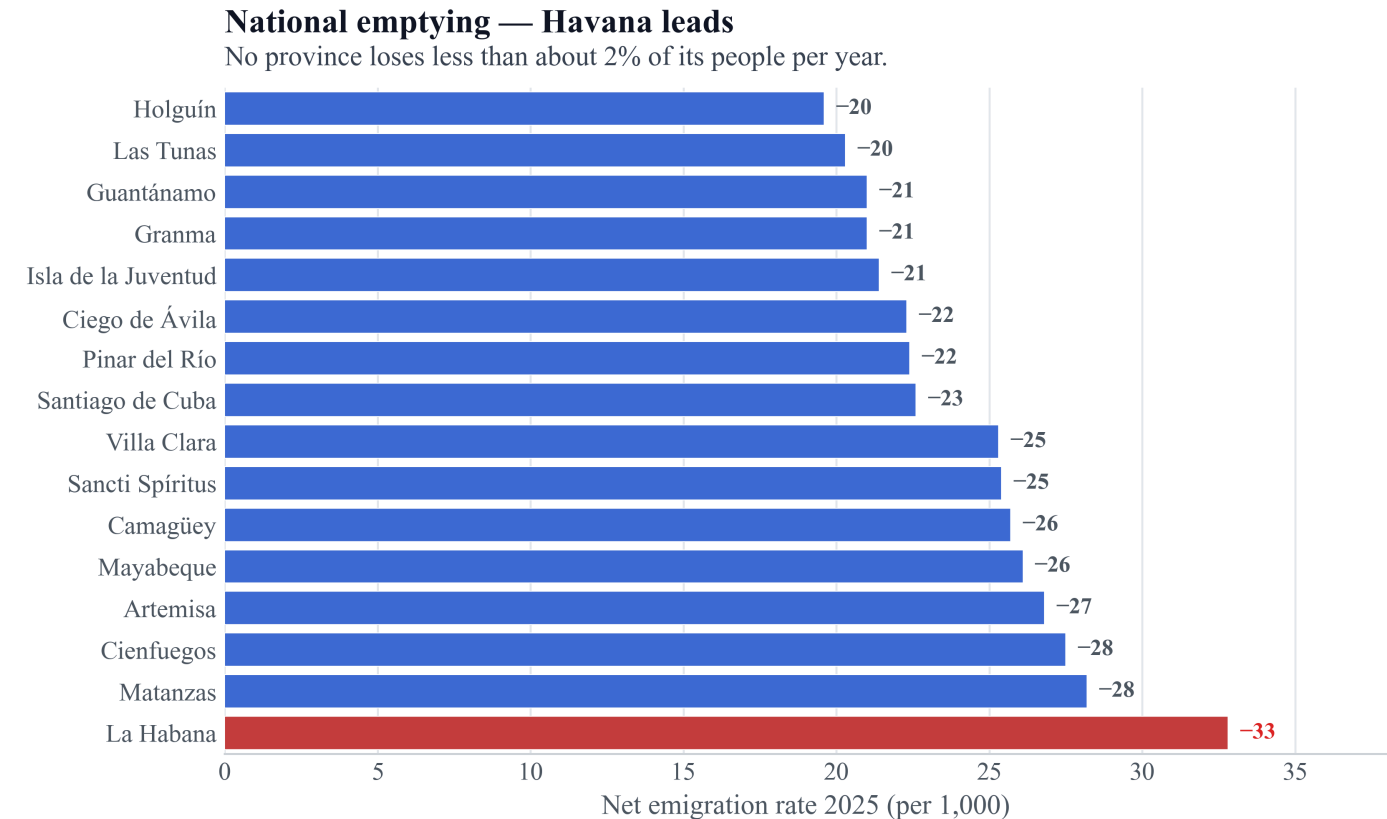
77% of emigrants are aged 15–59. Each young exit reduces future births.



Age composition of emigrants vs residents; median age leavers ≈30, stayers 45.

Emptying is national; Havana leads

All provinces lose about 2% of their people per year (ONEI; levels may be biased).



Net emigration 2025 per 1,000 inhabitants. Source: ONEI Indicators 2025.

HOW TO CITE

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METHOD

Demographic balance solved by Monte Carlo (2×10^6 draws). Scenarios conservative / central / upper; the central scenario is the illustrative central path. Destination ledger floor ~1.05 M (U.S. admin + 250k non-U.S.). The step to ~2.0 M net under the central scenario is a prior on ONEI migration, not destination microdata.

SOURCES

ONEI yearbooks and Indicators 2025. UN WPP. Albizu-Campos; Brønnum-Hansen & Albizu (2023). Data and code: github.com/ypriverol/cubascience